

Functional

- Users can submit a long URL and receive a unique shortened URL.
- Users can access the original URL by visiting the shortened URL.
- The system should redirect users to the original URL with minimal delay.
- Each generated short URL should be unique.
- Users can delete or deactivate an existing short URL .
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Non-Functional

- The system should provide **high availability** (e.g., 99.99% uptime).
- The system should support **high scalability** to handle millions of users and billions of URLs.
- URL redirection should have **low latency** (preferably under 100 ms).
- URL creation should be fast and efficient.
- The system should ensure **high reliability**, so every short URL always redirects to the correct destination.
- The system should provide **data durability**, ensuring URL mappings are not lost during failures.

Estimation

Assumption that we we are providing and how many users we have :

- Total Registered Users = **10 Million**
- Daily Active Users = **1 Million**
- New Short URLs Created per Day = **10 Million**
- URL Redirects per Day = **100 Million**
- Read : Write Ratio = **10 : 1**
- Average Original URL Length = **200 Bytes**
- Average Short URL Length = **8 Characters**
- Metadata (ID, timestamps, etc.) = **100 Bytes**
- Total Storage per URL ≈ **350 Bytes**

Requirements according to the estimation

Traffic

- Average Write Requests: 116 requests/sec
- Peak Write Requests: 350 requests/sec
- Average Read Requests: 1,157 requests/sec
- Peak Read Requests: 3,500 requests/sec
- Total Peak Requests: ~3,850 requests/sec

Storage

- Storage Per URL: 350 Bytes
- Daily Storage: 3.5 GB
- Monthly Storage: 105 GB
- Yearly Storage: 1.28 TB
- Storage for 5 Years: 6.4 TB
- Database Storage (including replicas & indexes): ~20 TB

Cache

- Cache Strategy: Store the most frequently accessed URLs
- Cache Size: 2–4 GB
- Cache Technology: Redis

- Application Servers: 6–8
- Load Balancers: 2
- Redis Cache Nodes: 2
- Database Servers:
- 1 Primary Database
- 2 Read Replicas

Database Requirements

- Total Database Storage: ~20 TB
- SSD Storage
- 64–128 GB RAM per Database Server

